

Statistical Process Control

1 Introduction to Control Charts

Uses of the control chart

- Most processes do not operate in a state of statistical control
- Consequently, an attentive use of control charts will identify assignable causes. If these causes can be eliminated from the process, variability will be reduced and the process will be improved
- The control chart only detects assignable causes. Management, operator, and engineering action will be necessary to eliminate the assignable causes.

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Introduction to Control Charts

Control charts kinds

- Variables Control Charts
 - The charts are applied to data that follow a continuous distribution
- Attributes Control Charts
 - The charts are applied to data that follow a discrete distribution

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3 Introduction to Control Charts

Control charts kinds

- Variable data
 - ✓ Product characteristic that can be measured:
 - ✓ Length, size, weight, height, time, velocity
 - ✓ Variable charts require smaller samples 2 to 10 parts in a sample
- Attribute data
 - ✓ Product characteristic evaluated with a discrete choice
 - ✓ Good/bad, yes/no
 - ✓ Attribute charts require larger sample sizes 50 to 100 parts in a sample

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4 Attribute Control Charts

- There are various types of control charts for attributes:
 - The p -chart for Proportion Defective
 - The np -chart for the number of defectives (or nonconforming units)
 - The c -chart for the number of defects
 - The u -chart for the number of defects per unit

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p-Chart

- ❑ p -chart is a process chart that is used to control the proportion of items in a sample that are defective.
- ❑ Samples are not necessary identical
- ❑ p -charts are effectively used to determine when there has been a shift in the proportion defective for a particular product or service.
- ❑ Examples of applications of the p -chart: incomplete orders, late deliveries, missed calls, accounting transaction errors, or parts that don't mate properly.

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p-Chart with constant sample size

Initial Control Limits

$$\bar{p} = \frac{\sum np}{\sum n}$$

$$UCL_p = \bar{p} + 3 \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n}}$$

$$LCL_p = \bar{p} - 3 \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n}}$$

Revised Control Limits

$$\bar{p}_{\text{new}} = \frac{\sum np - np_d}{\sum n - n_d}$$

$$UCL_p = \bar{p}_{\text{new}} + 3 \frac{\sqrt{\bar{p}_{\text{new}}(1-\bar{p}_{\text{new}})}}{\sqrt{n}}$$

$$LCL_p = \bar{p}_{\text{new}} - 3 \frac{\sqrt{\bar{p}_{\text{new}}(1-\bar{p}_{\text{new}})}}{\sqrt{n}}$$

All of the points should fall between the upper and lower control limits

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p-Chart with constant sample size

Sample size $n = 100$

Sample number	Number of errors	Proportion defective
1	6	6,00%
2	5	5,00%
3	0	0,00%
4	1	1,00%
5	4	4,00%
6	2	2,00%
7	5	5,00%
8	3	3,00%
9	3	3,00%
10	2	2,00%
11	6	6,00%
12	1	1,00%
13	8	8,00%
14	7	7,00%
15	5	5,00%
16	4	4,00%
17	11	11,00%
18	3	3,00%
19	0	0,00%
20	4	4,00%
	80	4,00%

$$\bar{p} = \frac{\sum n \times p_i}{\sum n} = \frac{80}{100 \times 20} = 0,04$$

$$UCL(p) = \bar{p} + 3 \frac{\sqrt{\bar{p}(1 - \bar{p})}}{\sqrt{n}} = 0,099 = 9,9\%$$

$$LCL(p) = \bar{p} - 3 \frac{\sqrt{\bar{p}(1 - \bar{p})}}{\sqrt{n}} = -0,019 = 0\%$$

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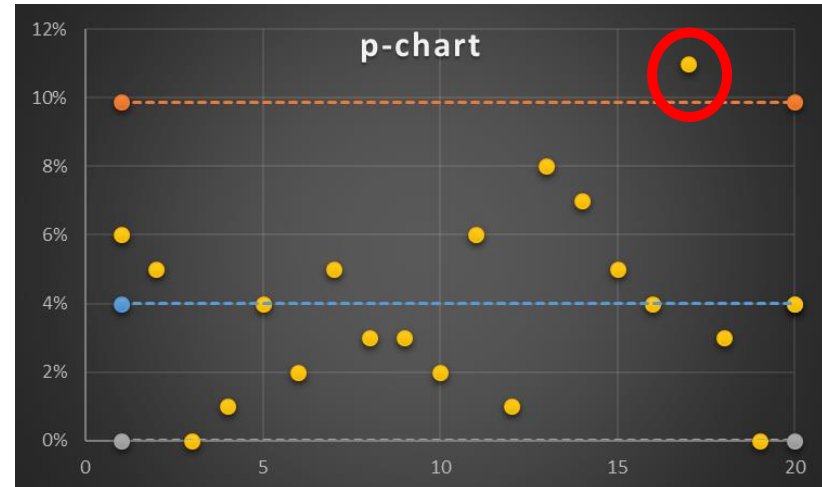
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p-Chart with constant sample size

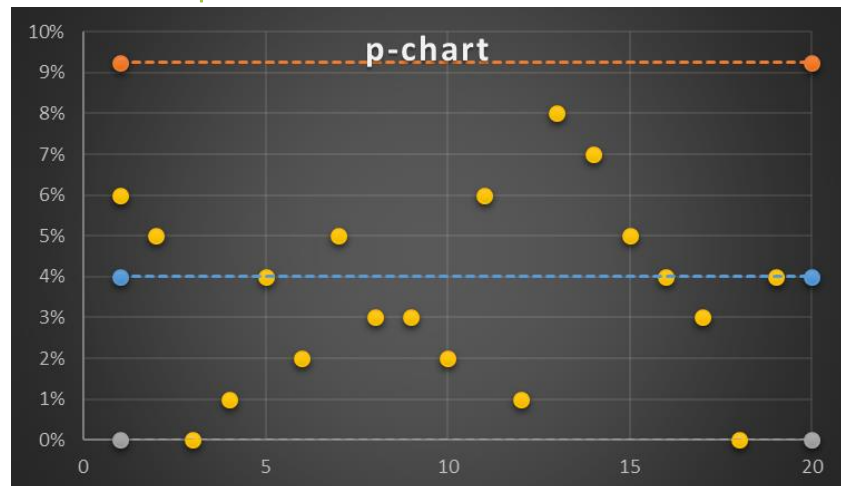
Sample size $n = 100$

Sample Number	Number of errors	Proportion defective
1	6	6,00%
2	5	5,00%
3	0	0,00%
4	1	1,00%
5	4	4,00%
6	2	2,00%
7	5	5,00%
8	3	3,00%
9	3	3,00%
10	2	2,00%
11	6	6,00%
12	1	1,00%
13	8	8,00%
14	7	7,00%
15	5	5,00%
16	4	4,00%
17	11	11,00%
18	3	3,00%
19	0	0,00%
20	4	4,00%
	80	4,00%

Initial p-chart



Revised p-chart



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p-Chart with variable sample size

- When the number of items sampled varies, the p -chart can be easily adapted to varying sample sizes.
- If the sample size varies
 - the control limits must be calculated for each different sample size by changing the value of n in the control limit formulas each time a different sample size is taken.
 - calculating the centerline and interpreting the chart will be the same

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10 p-Chart with variable sample size

- Calculate the centerline.
- For each sample calculate a separate pair of control limits.
- Let m = number of samples.

$$\bar{p} = \frac{\sum_{i=1}^m n_i p_i}{\sum_{i=1}^m n_i} \quad (\text{one centerline for all samples})$$

$$UCL_p = \bar{p} + 3 \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n_i}} \quad \text{for the } i\text{-th sample}$$

$$LCL_p = \bar{p} - 3 \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n_i}} \quad \text{for the } i\text{-th sample}$$

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p-Chart with variable sample size

- The calculation of control limits for the p chart with variable sample size can be simplified with the use of n_{ave}
- The value n_{ave} can be used whenever the individual sample sizes vary no more than 25% from the calculated n_{ave}

$$0,75 \times n_{ave} \leq n_i \leq 1,25 \times n_{ave}$$

- The value n_{ave} can be found by summing up the individual sample sizes and dividing by the total number of samples were taken:

$$n_{ave} = \frac{\sum_{i=1}^m n_i}{m}$$

where, m = number of samples

$$UCL_p = \bar{p} + 3 \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n_{ave}}}$$
$$LCL_p = \bar{p} - 3 \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n_{ave}}}$$

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12 p-Chart with variable sample size

Sample i	number of defective	sample size n_i	%P
1	9	256	3,516%
2	5	251	1,992%
3	10	248	4,032%
4	1	256	0,391%
5	6	260	2,308%
6	9	260	3,462%
7	6	250	2,400%
8	9	249	3,614%
9	6	255	2,353%
10	8	252	3,175%
p_bar	2,720%		

$$\bar{p} = 0,0272$$

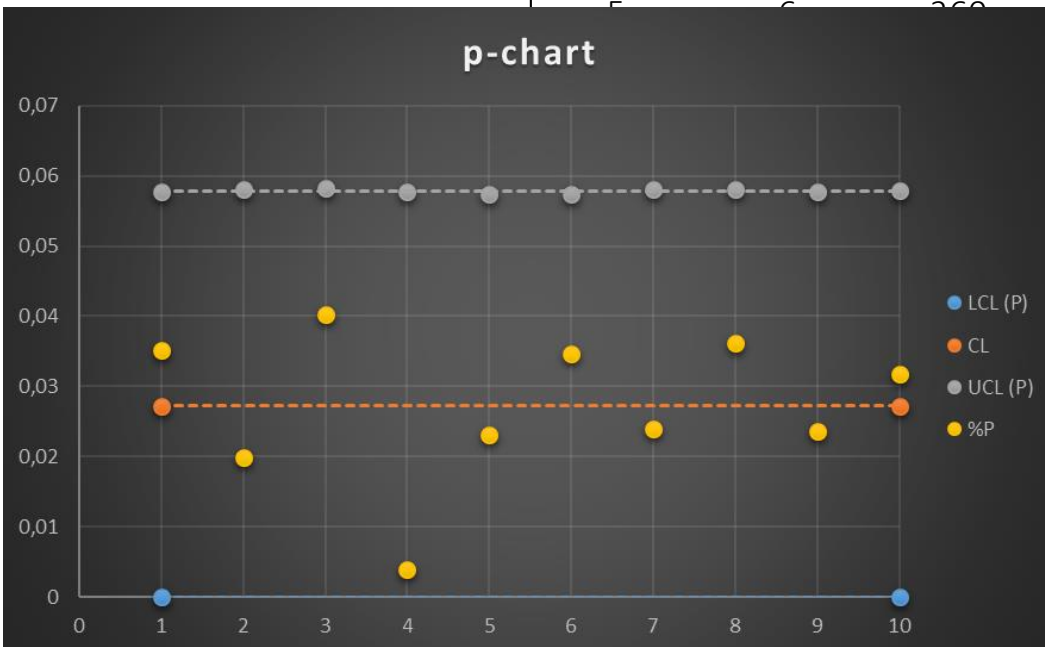
$$UCL(p) = \bar{p} + 3 \frac{\sqrt{\bar{p}(1 - \bar{p})}}{\sqrt{n_i}}$$

$$LCL(p) = \max \left(0, \bar{p} - 3 \frac{\sqrt{\bar{p}(1 - \bar{p})}}{\sqrt{n_i}} \right)$$

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13 p-Chart with variable sample size

Sample i	number of defective	sample size ni	%P	LCL (P)	CL	UCL (P)
1	9	256	3,516%	0	2,720%	5,77%
2	5	251	1,992%	0	2,720%	5,80%
3	10	248	4,032%	0	2,720%	5,82%
4	1	256	0,391%	0	2,720%	5,77%
5	6	260	2,308%	0	2,720%	5,75%
6	9	259	3,462%	0	2,720%	5,75%
7	6	254	2,400%	0	2,720%	5,81%
8	9	257	3,614%	0	2,720%	5,81%
9	5	253	2,353%	0	2,720%	5,78%
10	7	260	3,175%	0	2,720%	5,79%



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np-Chart

- The **np-chart** is a graph of the number of defectives (or nonconforming units) in a sample.
- The **np-chart** requires that the sample size of each subgroup be the same.
- When sample sizes are equal, either the **p** or **np-chart** can be used (they are essentially the same chart).
- Some people find the **np-chart** easier to use because it reflects integer numbers rather than proportions.

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np-Chart

- The **np-chart** construction steps are similar to those of the **p-chart**. The trial centerline and control limits are as follows:

$$\text{Centerline } n\bar{p} = \frac{\sum np}{m}$$

where, m = number of subgroups

$$UCL_{np} = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})}$$

$$LCL_{np} = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$$

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c-chart

Defective and defect

- A defective article is the one that fails to conform to some specification.
- Each instance of the article's lack of conformity to specifications is a defect
- A defective article may have one or more defects
- The c-chart applies to the total number of defects in samples of constant size

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c-chart

$$\bar{c} = \frac{\sum_i c_i}{m}$$

$$UCL(C) = \bar{c} + 3 \times \sqrt{\bar{c}}$$

$$LCL(C) = \bar{c} - 3 \times \sqrt{\bar{c}}$$

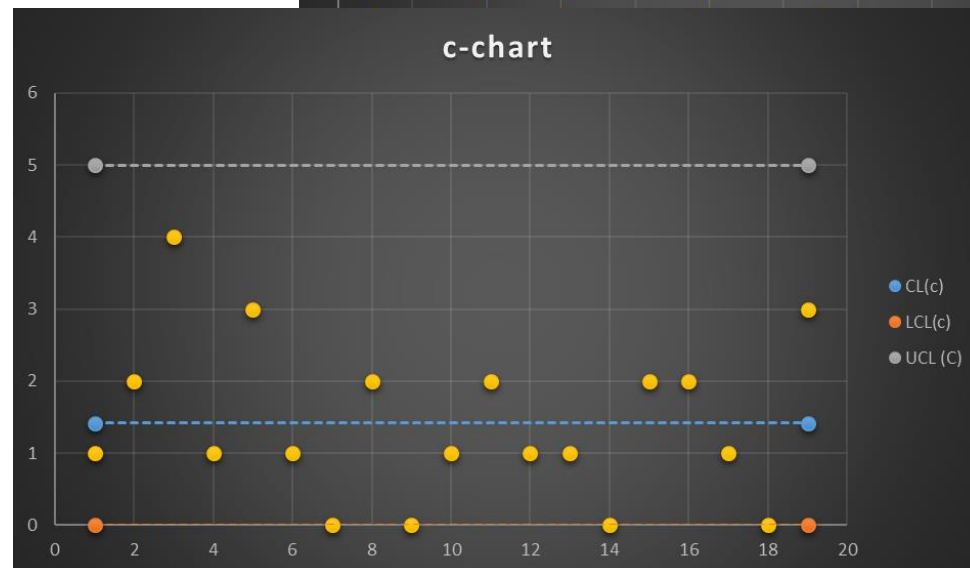
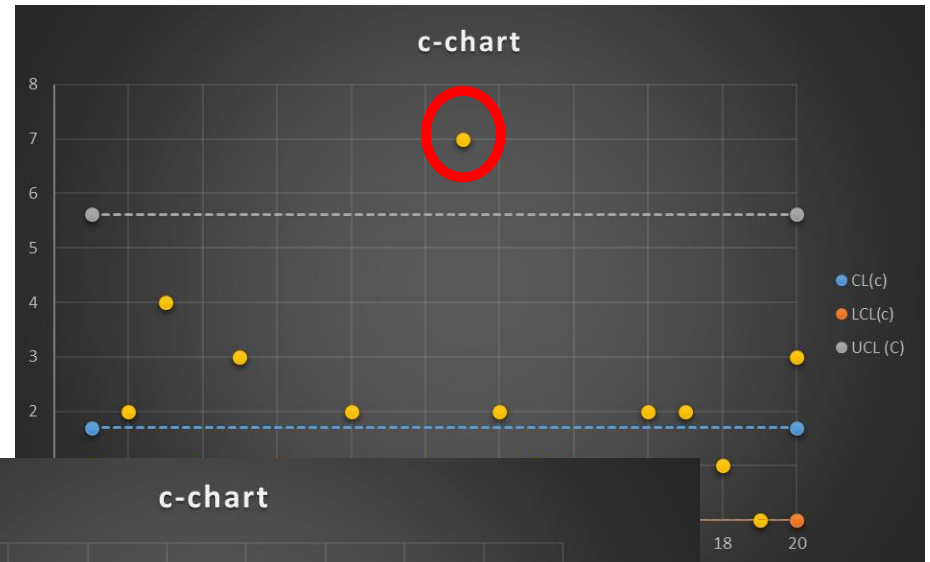
- c_i : the number of defects observed for the i^{th} controlled unit
- m is the number of samples
- Sample have same size

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c-chart

Sample N°	Number of defective
1	1
2	2
3	4
4	1
5	3
6	1
7	0
8	2
9	0
10	1
11	7
12	2
13	1
14	1
15	0
16	2
17	2
18	1
19	0
20	3
Sum	34



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u-chart

- The u-chart considers the average number of defects per unit.
- This is contrasted with the c-chart, which shows the actual number of defects per standardized unit.
- The u chart allows for the units sampled to be different sizes

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u-chart

- c_i : the number of defects observed for the i^{th} controlled sample unit n_i .
- The value of $u_i = \frac{c_i}{n_i}$
- The central value: $\bar{u} = \frac{\sum n_i \times u_i}{\sum n_i}$

$$\bar{u} = \frac{\sum_i n_i u_i}{\sum n_i}$$

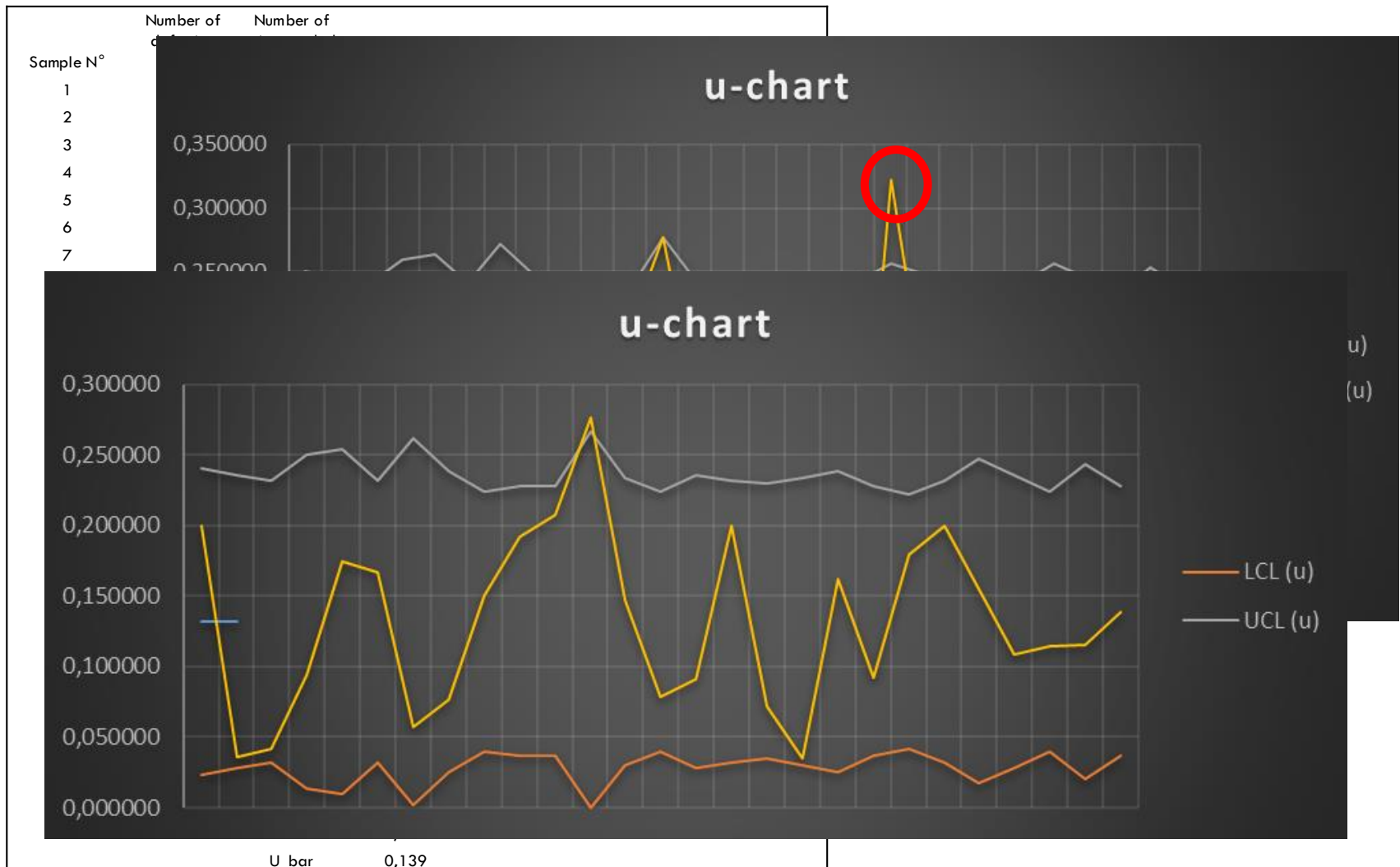
$$UCL(u) = \bar{u} + 3 \times \sqrt{\frac{\bar{u}}{n_i}}$$

$$LCL(u) = \bar{u} - 3 \times \sqrt{\frac{\bar{u}}{n_i}}$$

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u-chart



CUSUM Chart

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CUSUM Chart

- A limitation of the traditional $\bar{X}$ -chart is its inability to detect a relatively small change in a process mean.
- *Cumulative Sum (CUSUM)* control charts and procedures are designed to remedy this defect.
- Target value is subtracted from every control analyses and difference added to the sum of all previous differences
- There are two equivalent versions of a CUSUM chart: **one graphical and the other computational.**

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The V-Mask version

- Let μ_0 denote a target value or goal for the process mean and define cumulative sums by:

$$\begin{cases} s_1 = (\bar{x}_1 - \mu_0) \\ s_2 = (\bar{x}_1 - \mu_0) + (\bar{x}_2 - \mu_0) \\ \vdots \\ s_l = \sum_{k=1}^l (\bar{x}_k - \mu_0) \end{cases}$$

- The cumulative sums are plotted over time. At time l , we plot a point at height s_l .

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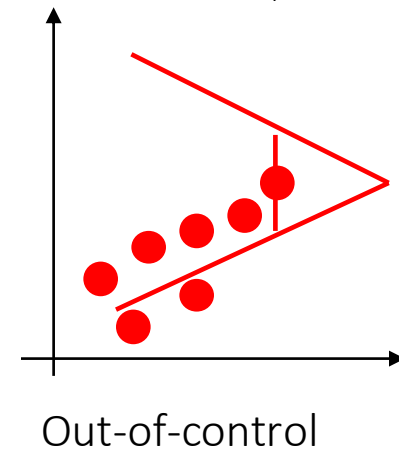
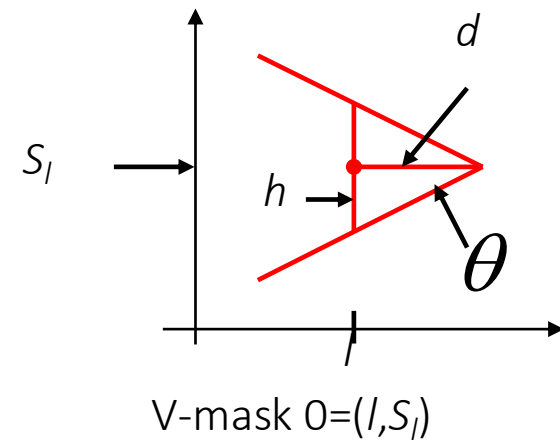
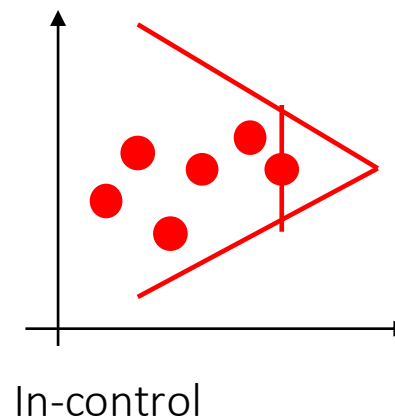
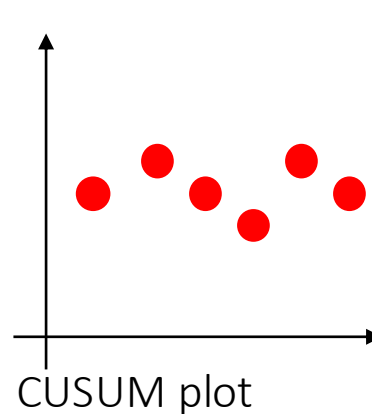
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The V-Mask version

- A V-shaped mask is superimposed on the plot. The process is judged to be out of control if any of the plotted points lies outside the V-mask.

Choose d and θ so that:

- Very few false alarms occur when the process is under control
- An important change in the process mean is quickly detected



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CUSUM chart: computational version

- Let x_i be the i th observation on the process
- If the process is in control then $x_i \rightarrow N(\mu_0, \sigma)$
- σ is known or can be estimated.
- Accumulate derivations from the target μ_0 above the target are noted by C^+
- Accumulate derivations from the target μ_0 below the target are noted by C^-
- C^+ and C^- are one-sided upper and lower CUSUMS, respectively.

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CUSUM chart: computational version

The parameters are computed as follows:

$$C_i^+ = \max \left[0, x_i - (\mu_0 + k) + C_{i-1}^+ \right]$$
$$C_i^- = \max \left[0, (\mu_0 - k) - x_i + C_{i-1}^- \right]$$

- Initial values are $C_0^+ = C_0^- = 0$
- k is the **reference value**
- If either C^+ or C^- exceed a decision interval **H** , the process is considered to be out of control.
- Often taken as a **$H = 5\sigma$**

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CUSUM chart: computational version

Selecting the reference value, k

- k is often chosen halfway between the target μ_0 and the out-of-control value of the mean μ_1 that we are interested in detecting quickly.
- Shift is expressed in standard deviation units as:

$$\mu_1 = \mu_0 + \delta\sigma$$

then

$$k = \frac{|\mu_1 - \mu_0|}{2} = \frac{\delta}{2} \sigma$$

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CUSUM chart: computational version

How to Determine if Process Out-of-Control?

- H - decision interval
- If C^+ or C^- exceed the decision interval (H), the process is considered out-of-control
- Rule of thumb value for H
 - ✓ Choose H to be five times the process standard deviation, $H = H = 5\sigma$
- Counters N^+ and N^- record the number of consecutive periods the CUSUM C^+ and C^- are above zero, respectively.
 - ✓ The counters can be used to indicate when the shift most likely occurred

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CUSUM chart: computational version (Example)

- Assume the process is in control then $x_i \rightarrow N(\mu_0 = 10, \sigma = 1)$ with sample size $n=1$.
- Say magnitude of shift we want to detect is $\delta = 1$
- So, we have

$$\begin{cases} k = \frac{\delta}{2} \sigma = \frac{1}{2} \times 1 = 0.5 \text{ (lower limit)} \\ H = 5 \times \sigma = 5 \text{ (upper limit)} \end{cases}$$

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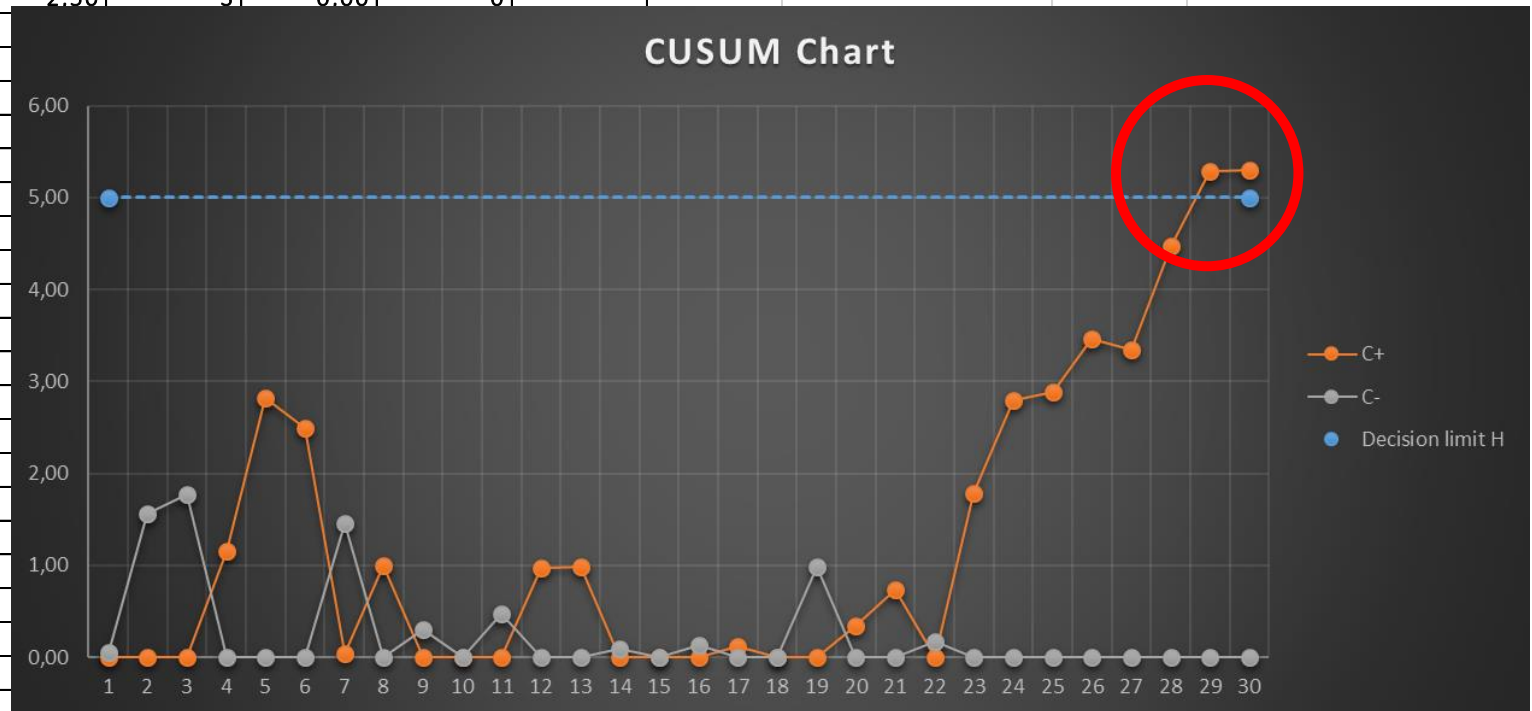
CUSUM chart: computational version (Example)

Period k	x_i	C_i^+	N_i^+	C_i^-	N_i^-	Decision
1	9,45	0,00	0	0,05	1	****
2	7,99	0,00	0	1,56	2	****
3	9,29	0,00	0	1,77	3	****
4	11,66	1,16	1	0,00	0	****
5	12,16	2,82	2	0,00	0	****
6	10,18	2,50	3	0,00	0	****
7	8,04					
8	11,46					
9	9,20					
10	10,34					
11	9,03					
12	11,47					
13	10,51					
14	9,40					
15	10,08					
16	9,37					
17	10,62					
18	10,31					
19	8,52					
20	10,84					
21	10,90					
22	9,33					
23	12,29					
24	11,50					
25	10,60					
26	11,08					
27	10,38					
28	11,62	4,47	6	0,00	0	****
29	11,31	5,28	7	0,00	0	Stop
30	10,52	5,30	8	0,00	0	Stop

Decision interval (H) 5.0

$$C_i^+ = \max \left[0, x_i - (\mu_0 + k) + C_{i-1}^+ \right]$$

$$C_i^- = \max \left[0, (\mu_0 - k) - x_i + C_{i-1}^- \right]$$



Control Chart Performance

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33 Control Chart Performance

- Assume the process is in control then $x_i \rightarrow N(\mu_0, \sigma_0)$
- Let consider magnitude of shift as follows :
$$\begin{cases} \mu_1 = \mu_0 + k\sigma_0 \\ \sigma_1 = \rho\sigma_0 \end{cases}$$
- Let p = probability that any point does not exceed the control limits.

Then,

$$p = F\left(\frac{3 - k\sqrt{n}}{\rho}\right) - F\left(\frac{-k\sqrt{n} - 3}{\rho}\right)$$

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34 Control Chart Performance

Average Run Length

The **average run length** (ARL) is a way of determining the appropriate sample size and sampling frequency.

$$ARL = \frac{1}{1 - p}$$

PS. this formula is based on $\sum_{j=0}^{+\infty} j \times x^{j-1} = \frac{1}{(1-x)^2}$

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35 Control Chart Performance

Average Run Length

The **average run length** (ARL) is a way of determining the appropriate sample size and sampling frequency.

	ARL	ARL	ARL
	n= 1	n=5	n=7
Magnitude of process shift			
(k= 0, ρ =1)	370.39	370.39	370.39
(k= 0.5, ρ =1)	155.22	33.40	21.38
(k= 1, ρ =1)	43.89	4.49	2.76
(k= 1.5, ρ =1)	14.97	1.56	1.20
(k= 2, ρ =1)	6.30	1.07	1.01